Experiment – 5

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(A)

A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.

For each registered user, calculate the confirmation rate, which is defined as:

```
Confirmation Rate = Number of Successful Confirmations
                    ----------------------------------
                    Total Confirmation Requests
```

If a user has never received any confirmation request, their confirmation rate should be considered **0**.

Display the user ID along with the confirmation rate rounded to **2 decimal places**. The result can be returned in any order.

Signups

user_id	time_stamp
3	2020-03-21 10:16:13
7	2020-01-04 13:57:59
2	2020-07-29 23:09:44
6	2020-12-09 10:39:37

Confirmations

user_id	time_stamp	action
3	2021-01-06 03:30:46	timeout
3	2021-07-14 14:00:00	timeout
7	2021-06-12 11:57:29	confirmed
7	2021-06-13 12:58:28	confirmed
7	2021-06-14 13:59:27	confirmed
2	2021-01-22 00:00:00	confirmed
2	2021-02-28 23:59:59	timeout

Output

user_id	confirmation_rate
2	0.50
3	0.00
6	0.00
7	1.00

Solution:

```
CREATE TABLE Signups
(
    user_id INT PRIMARY KEY,
    time_stamp TIMESTAMP
);

CREATE TABLE Confirmations
(
    user_id INT,
    time_stamp TIMESTAMP,
    action VARCHAR(20),
    PRIMARY KEY(user_id, time_stamp),
    FOREIGN KEY(user_id)
    REFERENCES Signups(user_id)
);

INSERT INTO Signups
VALUES
(3,'2020-03-21 10:16:13'),
(7,'2020-01-04 13:57:59'),
(2,'2020-07-29 23:09:44'),
(6,'2020-12-09 10:39:37');

INSERT INTO Confirmations
VALUES
(3,'2021-01-06 03:30:46','timeout'),
(3,'2021-07-14 14:00:00','timeout'),
(7,'2021-06-12 11:57:29','confirmed'),
(7,'2021-06-13 12:58:28','confirmed'),
(7,'2021-06-14 13:59:27','confirmed'),
(2,'2021-01-22 00:00:00','confirmed'),
(2,'2021-02-28 23:59:59','timeout');

SELECT S.USER_ID,
      ROUND(
            AVG(
                  CASE
                        WHEN ACTION = 'confirmed' THEN 1
                        ELSE 0
                  END
            ), 2
      ) AS CONFIRMATION_RATE
FROM SIGNUPS AS S
LEFT JOIN CONFIRMATIONS AS C
ON S.USER_ID = C.USER_ID
GROUP BY S.USER_ID;
```

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(B)

A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between **2013-10-01** and **2013-10-03**, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

We need to calculate:

Cancellation Rate

=

Cancelled Trips

Total Valid Trips

But only consider trips where:

Client is NOT banned
AND
Driver is NOT banned

and

Date between

2013-10-01
And
2013-10-03

Users

users_id	banned	role
1	No	client
2	Yes	client
3	No	client
4	No	client
10	No	driver
11	No	driver
12	No	driver
13	No	driver

Trips

id	client_id	driver_id	city_id	status	request_at
1	1	10	1	completed	2013-10-01
2	2	11	1	cancelled_by_driver	2013-10-01
3	3	12	6	completed	2013-10-01
4	4	13	6	cancelled_by_client	2013-10-01
5	1	10	1	completed	2013-10-02
6	2	11	6	completed	2013-10-02
7	3	12	6	completed	2013-10-02
8	2	12	12	completed	2013-10-03
9	3	10	12	completed	2013-10-03
10	4	13	12	cancelled_by_driver	2013-10-03

Output

Day	Cancellation Rate
2013-10-01	0.33
2013-10-02	0.00
2013-10-03	0.50

Solution:

```
CREATE TABLE Users
(
    users_id INT PRIMARY KEY,
    banned VARCHAR(3),
    role VARCHAR(20)
);

CREATE TABLE Trips
(
    id INT PRIMARY KEY,
    client_id INT,
    driver_id INT,
    city_id INT,
    status VARCHAR(30),
    request_at DATE
);

INSERT INTO Users
VALUES
(1,'No','client'),
(2,'Yes','client'),
(3,'No','client'),
(4,'No','client'),
(10,'No','driver'),
(11,'No','driver'),
(12,'No','driver'),
(13,'No','driver');

INSERT INTO Trips
VALUES
(1,1,10,1,'completed','2013-10-01'),
(2,2,11,1,'cancelled_by_driver','2013-10-01'),
(3,3,12,6,'completed','2013-10-01'),
(4,4,13,6,'cancelled_by_client','2013-10-01'),
(5,1,10,1,'completed','2013-10-02'),
(6,2,11,6,'completed','2013-10-02'),
(7,3,12,6,'completed','2013-10-02'),
(8,2,12,12,'completed','2013-10-03'),
(9,3,10,12,'completed','2013-10-03'),
(10,4,13,12,'cancelled_by_driver','2013-10-03');
```

```
SELECT T.REQUEST_AT AS DAY,
      ROUND(
            AVG(
                  CASE
                        WHEN T.STATUS IN ('cancelled_by_driver',
                        'cancelled_by_client') THEN 1
                        ELSE 0
                  END
            ), 2
      ) AS CANCELLATION_RATE
FROM TRIPS AS T
JOIN USERS AS C
ON T.CLIENT_ID = C.USERS_ID
JOIN USERS AS D
ON T.DRIVER_ID = D.USERS_ID
WHERE C.BANNED = 'No'
AND D.BANNED = 'No'
AND T.REQUEST_AT BETWEEN '2013-10-01' AND '2013-10-03'
GROUP BY T.REQUEST_AT;
```